

Answer Specimen SM015

NO	ANSWER SCHEME	MARKS	REMARKS
1.	(a) $\lim_{x \rightarrow \infty} \frac{2x^2 + 8x - 2}{x^2 + 1}$ $= \lim_{x \rightarrow \infty} \frac{2x^2/x^2 - 8x/x^2 - 2/x^2}{x^2/x^2 + 1/x^2}$ $= \lim_{x \rightarrow \infty} \frac{2 + 8/x - 2/x^2}{1 + 1/x^2}$ $= \frac{2 + 0 - 0}{1 + 0} = 2$	B1 M1 A1	Divide with the highest power of x at denominator Simplify As it is
		3 marks	
	(b) Continuous at x = 1 $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$ $-4 = 4(1)^2 - 5H(1) - K$ $-4 = 4 - 5H - K$ $-8 = -5H - K$ $8 = 5H + K$ ----- (1) Continuous at x = -1 $\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x) = f(-1)$ $4(1)^2 - 5H(-1) - K = H(-1) + K$ $4 + 5H - K = -H + K$ $4 = -6H + 2K$ $2 = -3H + K$ ----- (2) $6 = 8H$ $H = \frac{3}{4}$ $2 = -3(\frac{3}{4}) + K$ $K = \frac{17}{4}$	M1 M1 M1 M1 A1 A1	Simplify First equation Simplify Second equation As it is for H As it is for K
		6 marks	

2.	$\text{a) } f'(x) = \lim_{h \rightarrow 0} \frac{\frac{2}{3\sqrt{x+h}} - \frac{2}{3\sqrt{x}}}{h}$ $f'(x) = \lim_{h \rightarrow 0} \frac{2}{3} \left(\frac{\sqrt{x} - \sqrt{x+h}}{(\sqrt{x})(\sqrt{x+h})} \right) \times \frac{1}{h}$ $f'(x) = \lim_{h \rightarrow 0} \left(\frac{2}{3} \right) \frac{(\sqrt{x} - \sqrt{x+h})}{(\sqrt{x+h})(\sqrt{x})} \times \frac{(\sqrt{x} + \sqrt{x+h})}{(\sqrt{x} + \sqrt{x+h})} \times \frac{1}{h}$ $f'(x) = \lim_{h \rightarrow 0} \left(\frac{2}{3} \right) \frac{(x - (x+h))}{(\sqrt{x+h})(\sqrt{x})(\sqrt{x} + \sqrt{x+h})} \times \left(\frac{1}{h} \right)$ $f'(x) = \lim_{h \rightarrow 0} \left(\frac{2}{3} \right) \frac{-1}{(\sqrt{x+h})(\sqrt{x})(\sqrt{x} + \sqrt{x+h})}$ $f'(x) = -\frac{1}{3x^{3/2}}$	B1 M1 M1 M1 A1	Use the formulae correctly Simplify (Single fraction) Multiply with conjugate Simplify As it is
		5 marks	
	b) $x^2 y - \sin(2x^2) = 0$ $x^2 \frac{dy}{dx} + y 2x - \cos(2x^2)(4x) = 0$ $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + 2y + 2x \frac{dy}{dx} - [\cos(2x^2)(4) + 4x(-\sin(2x^2))(4x)] = 0$ $x^2 y'' + 2xy' + 2y + 2xy' - 4\cos(2x^2) + 16x^2 \sin(2x^2) = 0$ $x^2 y'' + 4xy' + 2y - 4\cos(2x^2) + 16x^2 x^2 y = 0$ $x^2 y'' + 4xy' + 2y - 4\cos(2x^2) + 16x^4 y = 0$ $x^2 y'' + 4xy' + 2y(1 + 8x^4) - 4\cos(2x^2) = 0 \text{ (proven)}$ $2y(1 + 8x^4) + 4xy' + x^2 y'' - 4\cos(2x^2) = 0$	M1 M1M1 M1 M1 A1	Use product rule Use product rule 2 times Simplify Substitute
		6 marks	
3.	$\frac{dV}{dt} = 80\pi$ $V = 4\pi h(h + 4)$ $= 4\pi h^2 + 16\pi h$ $\frac{dV}{dh} = 8\pi h + 16\pi$	M1	Differentiation

	$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$	B1	Rate of change
	$= \frac{1}{8\pi h + 16\pi} \times 80\pi$	M1	Simplify
	$= \frac{20}{2\pi h + 4\pi}$		
	When $h = 6 \text{ cm}$	M1	Substitute
	$\frac{dh}{dt} = \frac{10}{8} = \frac{5}{4} \text{ cms}^{-1}$	A1	As it is
		5 marks	

NO	ANSWER SCHEME	MARKS	REMARKS
1.	$1. \quad z = 2 - i \text{ and } w = \frac{i}{1-i}$ $z + w = 2 - i + \frac{i}{1-i}$ $= 2 - i + \frac{i(1+i)}{(1-i)(1+i)}$ $= 2 - i + \frac{i-1}{2}$ $= \left(2 - \frac{1}{2}\right) + \left(\frac{1}{2} - 1\right)i$ $= \frac{3}{2} - \frac{1}{2}i$ $ z + w = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{-1}{2}\right)^2}$ $= \frac{\sqrt{10}}{2}$ $\text{Arg}(z+w) = -\tan^{-1}\left(\frac{1}{3}\right) = -0.3218 \text{ rad}$ $z + w = \frac{\sqrt{10}}{2} [\cos(-0.3218) + i\sin(-0.3218)]$ $= \frac{\sqrt{10}}{2} [\cos(0.3218) - i\sin(0.3218)]$	B1 M1 M1 A1 M1 A1 A1 A1	Substitute Conjugate Simplify
		8 marks	
2	(a) $\frac{x+1}{x-2} - 3 \geq 0$ $\frac{x+1-3x+6}{x-2} \geq 0$ $\frac{-2x+7}{x-2} \geq 0$ i) $-2x+7 \geq 0 \quad , \quad x \leq \frac{7}{2}$ ii) $x-2 > 0 \quad , \quad x > 2$ Using number line The solution is $\left(2, \frac{7}{2}\right]$ (b) $\left \frac{x}{4+x}\right < 2$ $\frac{x}{4+x} < 2 \quad \text{AND} \quad \frac{x}{4+x} > -2$ $\frac{-x-8}{x+4} < 0 \quad \cap \quad \frac{3x+8}{x+4} > 0$	M1 M1 M1 A1 M1 M1	Rearrange Simplify Definition Either one

	i) $-x-8 > 0$, $x < -8$ i) $3x+8 > 0$, $x > -\frac{8}{3}$ ii) $x+4 > 0$, $x > -4$ ii) $x+4 > 0$, $x > -4$ $(-\infty, -8) \cup (-4, \infty)$ AND $(-\infty, 4) \cup \left(-\frac{8}{3}, \infty\right)$ Intersection between two sections The solution set is $(-\infty, -8) \cup \left(-\frac{8}{3}, \infty\right)$	M1 M1 M1 A1	
		10 marks	
3	(a) let $p^{-1}(x) = y \Rightarrow p(y) = x$ $e^{-2y} = x$ $\ln e^{-2y} = \ln x$ $-2y = \ln x$ $y = \frac{-\ln x}{2} \quad \therefore p^{-1}(x) = \frac{-\ln x}{2}$ (b) $(q \circ p)(x) = q(e^{-2x})$ $= 1 + (e^{-2x})^3$ $= 1 + e^{-6x}$ (c) $(p \circ q)(x) = e$ $p(q(x)) = e \quad -2(1+x^3) = 1$ $p(1+x^3) = e \quad x^3 = -1.5$ $e^{-2(1+x^3)} = e \quad x = -1.145$	M1 A1 M1 A1 M1 M1 A1	Taking ln Substitute Equating the index
		7 marks	
4	(a) (i) $\lim_{x \rightarrow 0^+} e^x = 1$ $\lim_{x \rightarrow 0^-} 1 - 9x^2 = 1$ $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$ $\therefore \lim_{x \rightarrow 0} f(x) = 1$ (ii) (a) $\therefore \lim_{x \rightarrow 0} f(x) = 1$ (b) $f(0) = e^0 = 1$ (c) $\therefore \lim_{x \rightarrow 0} f(x) = f(0) = 1$ So the function $f(x)$ is continuous at $x = 0$	M1 M1A1 M1 M1 A1	Step (a) Step (c)

	(b) $f(2) = 2(2)^2 + 8$ $= 16$ $\lim_{x \rightarrow 2^+} 4px = 8p$ Since , the function $f(x)$ is continuous at $x = 2$ $\lim_{x \rightarrow 2^+} 4px = f(2)$ $8p = 16$ $p = 2$	M1 M1 M1 M1A1	
		11 marks	
5	(a) $y = \ln\left(\frac{3+x}{3-x}\right)^{\frac{1}{2}}$ $y = \frac{1}{2} \ln\left(\frac{3+x}{3-x}\right)$ $= \frac{1}{2} [\ln(3+x) - \ln(3-x)]$ $\frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{3+x} + \frac{1}{3-x} \right]$ $= \frac{3}{(3+x)(3-x)}$ let $u = 3 - x$ $y = -\ln u$ and $\frac{dy}{du} = \frac{-1}{u}, \frac{du}{dx} = -1$ b) when $x = 1$, $(1)^2 + 4y^2 - 5y = 0$	M1 M1 M1A1 A1 M1	Use rule of log Substitute

	$(4y - 1)(y - 1) = 0$ $y = \frac{1}{4} \text{ and } y = 1$ $x^2 + 4y^2 - 5y = 0$ $2x + 8y \frac{dy}{dx} - 5 \left[y + x \frac{dy}{dx} \right] = 0$ $\frac{dy}{dx} [8y - 5x] = 5y - 2x$ $\frac{dy}{dx} = \frac{5y - 2x}{8y - 5x}$ $\text{when } x = 1, y = \frac{1}{4} \quad \text{when } x = 1 \text{ and } y = 1$ $\frac{dy}{dx} = \frac{5(\frac{1}{4}) - 2(1)}{8(\frac{1}{4}) - 5(1)} \quad \frac{dy}{dx} = \frac{5(1) - 2(1)}{8(1) - 5(1)}$ $= \frac{1}{4} \quad = 1$	A1 M1 M1 M1 A1 A1	Differentiate In terms of x and y Either one correct Both correct
		12 marks	
6	(a) $y = \frac{t^2}{t+1} \quad x = \frac{1}{t+1}$ $\frac{dy}{dt} = \frac{(t+1)2t - t^2}{(t+1)^2} \quad \frac{dx}{dt} = \frac{(t+1)0 - 1}{(t+1)^2}$ $= \frac{t^2 + 2t}{(t+1)^2} \quad = \frac{-1}{(t+1)^2}$ $= \frac{t(t+2)}{(t+1)^2}$ $\frac{dy}{dx} = \frac{t(t+2)}{(t+1)^2} \times \frac{(t+1)^2}{-1} = -t^2 - 2t$	M1M1 A1A1 M1A1	

	(b) $\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx}$ $= \frac{d}{dt} (-t^2 - 2t) \left(-(t+1)^2 \right)$ $= (-2t - 2) \left(-(t+1)^2 \right)$ $= 2(t+1)^3$ At the point $\left(\frac{1}{2}, \frac{1}{2} \right)$, $\frac{1}{t+1} = \frac{1}{2}$, $t=1$ $\frac{d^2y}{dx^2} = 2(1+1)^3 = 16$	M1 M1 A1 B1 M1A1	
		12	
7	(a) $f(x) = x + \frac{9}{x}$ $f'(x) = 1 - \frac{9}{x^2}, \quad f'(x) = 0$ $1 - \frac{9}{x^2} = 0$ $x^2 = 9, \quad x = \pm 3$ $f''(x) = \frac{18}{x^3}, \quad \text{when } x = 3, \quad f''(3) = \frac{2}{3} > 0 \text{ minimum}$ $f(3) = 6, \quad (3, 6) \text{ is a relative minimum}$ $f''(x) = \frac{18}{x^3}, \quad \text{when } x = -3, \quad f''(-3) = -\frac{2}{3} > 0 \text{ maximum}$ $f(-3) = -6, \quad (-3, -6) \text{ is a relative maximum}$ (b) (i) Let h = height of the oil tank V = volume of the oil tank Total surface area = 96 $2x^2 + 4xh = 96$ $x^2 + 2xh = 48$ $2xh = 48 - x^2$ $\therefore h = \frac{24}{x} - \frac{1}{2}x$	M1 A1 M1 A1 B1 M1	

	$v = x^2 h$ $(ii) \quad = x^2 \left(\frac{24}{x} - \frac{1}{2} x \right)$ $v = 24x - \frac{1}{2} x^3$ $\frac{dv}{dx} = 24 - \frac{3}{2} x^2$ $\frac{dv}{dx} = 0, \quad 24 - \frac{3}{2} x^2 = 0$ $x^2 = 16$ $x = 4$ $\frac{d^2v}{dx^2} = -3x$ when $x = 4$, $\frac{d^2v}{dx^2} = -3(4) = -12 < 0$ maximum $v = 24(4) - \frac{1}{2}(4)^3$ $= 64$ $\therefore$ the maximum volume of the oil tank is 64 m^2	M1 M1 A1 M1 M1 A1	V in single term
		15	